

光學 第一講

§2. 例1: 已知 y, n, i, j, y'

$$n \sin i = n' \sin i'$$

$$y' = \frac{\tan i}{\tan i'} y = \frac{\tan i \cdot y}{n \sin i} = \frac{y \cdot \sqrt{1-n'^2 \sin^2 i}}{n \cos i}$$

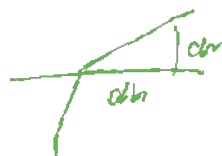


* 視像公式: 自左向右 $y' \approx \frac{y}{n}$

例3: $n^2 \cos^2 \theta = n_0^2 (1 - a^2 r^2)$ 求物面 $r(h)$

$$n \cos \theta = \frac{dh}{\sqrt{(dr)^2 + (dh)^2}}$$

$$= n \cos \theta \frac{1}{1 + \left(\frac{dr}{dh}\right)^2} = \frac{n_0 \sqrt{1 - a^2 r^2}}{1 + \left(\frac{dr}{dh}\right)^2} = \text{const.} = n_0 \frac{1}{1 + \left(\frac{dr}{dh}\right)^2} \Big|_{r=0}$$



$$\Rightarrow \frac{dr}{dh} = \sqrt{(1 - a^2 r^2) \left[1 + \left(\frac{dr}{dh}\right)_{r=0} \right] - 1} = \sqrt{-a^2 (k+1) r^2 + k} \quad \text{其中 } k = \left(\frac{dr}{dh}\right)^2 \Big|_{r=0}$$

$$h = \int \frac{dr}{\sqrt{-a^2 (k+1) r^2 + k}} = \frac{1}{\sqrt{k}} \frac{1}{a \sqrt{k+1}} \int \frac{d\left(\frac{a \sqrt{k+1}}{\sqrt{k}} r\right)}{\sqrt{1 - \frac{a^2 (k+1)}{k} r^2}}$$

$$= \frac{1}{a \sqrt{k+1}} \int \frac{d(\sin u)}{\cos u} = \frac{1}{a \sqrt{k+1}} \arcsin u \quad \text{其中 } \sin u = \frac{a \sqrt{k+1}}{\sqrt{k}} r$$

$$\text{得 } r = \frac{\sqrt{k}}{a \sqrt{k+1}} \sin(a \sqrt{k+1} h)$$

求解: 找 $n(r) \sim r(h)$

$$r = \frac{1}{a} \frac{dr}{dh}$$

$n \cos \theta = \text{const.}$

$$\cos \theta \frac{dn}{d\theta} + n \cdot (-\sin \theta) = 0 \Rightarrow \frac{dn}{d\theta} = n \tan \theta$$

$$\frac{dr}{dh} = \tan \theta \Rightarrow \frac{dn}{d\theta} = n \frac{dr}{dh} \quad \text{得 } \frac{dn}{d\theta} = n \cdot \frac{dr}{dh}$$

$$\frac{d^2 r}{dh^2} = \frac{1}{\cos^2 \theta} \frac{d\theta}{dh} = \frac{1}{n \cos^2 \theta} \frac{dn}{dr} = \frac{n}{n^2 \cos^2 \theta} \frac{dn}{dr} = \frac{1}{2 n^2 \cos^2 \theta} \frac{d(n^2)}{dr}$$

$$\text{得 } n \cos \theta = n_0 \cos \theta_0$$

$$\text{求解: } \frac{d^2 r}{dh^2} - k r = 0$$

$$- \text{解地物面: } \frac{d^2 r}{dh^2} - \frac{1}{2 n^2 \cos^2 \theta} \frac{d(n^2)}{dr} = 0$$

最小偏向角:

$$\begin{cases} \sin i_1 = n \sin i_1' \\ n \sin i_2 = \sin i_2' \end{cases}$$

$$i_1' + i_2 = \alpha$$

$$\begin{aligned} \delta &= i_1 + i_2' - i_1' - i_2' \\ &= i_1 + i_2' - \alpha \end{aligned}$$

$$\frac{d\delta}{di_1} = 1 + \frac{di_2'}{di_1} = 1 + \frac{di_2'}{di_2} \cdot \frac{di_2}{di_1} = 0$$

$$\frac{di_2'}{di_2} = \frac{n \cos i_2}{\cos i_2'} \quad \frac{di_2}{di_1} = -1, \quad \frac{di_1'}{di_1} = \frac{\cos i_1}{n \cos i_1'}$$

$$\Rightarrow \frac{d\delta}{di_1} = 1 - \frac{\cos i_1 \cos i_2}{\cos i_1' \cos i_2'} = 0 \Rightarrow \cos i_1 \cos i_2 = \cos i_1' \cos i_2'$$

$$\text{即: } \frac{\cos i_1}{1 - (\frac{\sin i_1}{n})^2} = \frac{\cos i_2'}{1 - (\frac{\sin i_2'}{n})^2} \quad i_1 = i_2' \text{ 或 } i_2 = i_1'$$

$$\Rightarrow \delta = \delta_{\min}$$

例 10: 用光路可逆证明上述结论.

惠更斯原理: 以波前的包络面是波前,

推导斯涅尔定律:

“入”为入射波前 “反”为反射波前

由同一波前等时: $t = t$ (入, 反射)

$$t = t \quad (\text{反、折射})$$

$$\Rightarrow i_1 = i_1', \quad \frac{\sin i_1}{v_1} = \frac{\sin i_2}{v_2} \quad \text{即} \quad n_1 \sin i_1 = n_2 \sin i_2$$

费马原理: $\delta L = \delta \int_L n dl = 0$

推导折射定律:

$$L = n_1 \frac{h_1}{\cos i} + n_2 \frac{h_2}{\cos i'}$$

$$\frac{dL}{di} = \frac{n_1 h_1 \sin i}{\cos^2 i} + \frac{n_2 h_2 \sin i'}{\cos^2 i'} \cdot \frac{di'}{di}$$

$$\left. \begin{matrix} n_1 \\ n_2 \end{matrix} \right\} \begin{matrix} h_1 \\ h_2 \end{matrix} \quad \begin{matrix} i \\ i' \end{matrix}$$

由约束: $h_1 \tan i + h_2 \tan i' = l$

对 i 求导, $h_1 \frac{1}{\cos^2 i} + h_2 \frac{1}{\cos^2 i'} \frac{di'}{di} = 0 \rightarrow \frac{di'}{di} = - \frac{\cos^2 i'}{\cos^2 i} \frac{h_1}{h_2}$

$$\frac{dL}{di} = \frac{n_1 h_1 \sin i}{\cos^2 i} - \frac{n_2 h_2 \sin i'}{\cos^2 i'} \cdot \frac{\cos^2 i'}{\cos^2 i} \frac{h_1}{h_2}$$

$$= \frac{h_1}{\cos^2 i} (n_1 \sin i - n_2 \sin i') = 0$$

$$\Rightarrow n_1 \sin i = n_2 \sin i'$$

光波为:

ψ : 辐照度

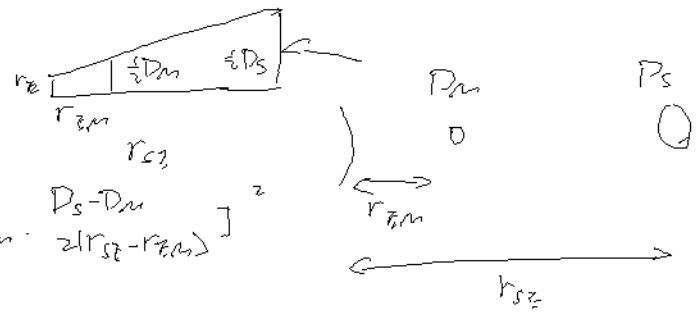
辐射通量, $\phi = \int \psi d\Omega dA$

2) 解:

1.1

$$S = \pi r_e^2$$

$$= \pi \cdot \left[\frac{1}{2} D_m - r_{em} \cdot \frac{D_s - D_m}{2(r_{se} - r_{em})} \right]^2$$



1.2

$$\vec{r} = x \cdot \vec{x} + y \cdot \vec{y} + z \cdot \vec{z}$$

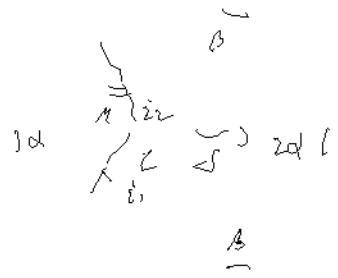
$$\text{经反射 } \vec{r}' = -x \cdot \vec{x} + (-y) \vec{y} + z \cdot \vec{z}$$

1.3

(1) 证明偏向角与入射方向无关且为 2α

$$\vec{r}_1 - \vec{i}_1 + \vec{r}_2 - \vec{i}_2 + \alpha = \vec{r}_1 \Rightarrow \vec{i}_1 + \vec{i}_2 = \alpha$$

$$\Delta = 2\vec{i}_1 + 2\vec{i}_2 = 2\alpha$$



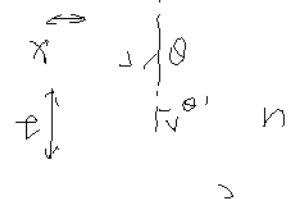
的色散? 否

1.4

试证光线通过平行板时的侧向位移为 $x = \frac{n-1}{n} \theta t$ ($\theta < 1$)

$$\sin \theta = n \sin \theta'$$

$$x^2 + \left(\frac{t}{\cos \theta'} \right)^2 - 2x \cdot \frac{t}{\cos \theta'} \cos(\frac{\pi}{2} + \theta') = \left(\frac{t}{\cos \theta} \right)^2$$



$$\Rightarrow x^2 + \frac{2t \sin \theta'}{\cos \theta'} x + t^2 \left(\frac{1}{\cos^2 \theta'} - \frac{1}{\cos^2 \theta} \right) = 0$$

$$\theta < 1 \Rightarrow \theta' < 1, \sin \theta \approx \theta, \sin \theta' \approx \theta' \Rightarrow \theta' = \frac{\theta}{n}$$

$$\Rightarrow x^2 + 2t \cdot \left(\frac{\theta}{n} \right) x + \frac{[\cos \theta + \cos \theta'] (\cos^2 \theta - \cos^2 \theta')}{\cos^2 \theta' \cos^2 \theta}$$

$$= x^2 + 2t \left(\frac{\theta}{n} \right) x - \frac{[2 - \frac{1}{2} \theta^2 (1 + \frac{1}{n^2})] \frac{\theta^2}{2} (1 - \frac{1}{n^2})}{[1 - \frac{\theta^2}{2n^2}] [1 - \frac{\theta^2}{2}]} t^2$$

$$\approx x^2 + 2t \left(\frac{\theta}{n} \right) x - \theta^2 \left(1 - \frac{1}{n^2} \right) t^2 = 0$$

$$\Delta = 4t^2 \frac{\theta^2}{n^2} + 4\theta^2 \left(1 - \frac{1}{n^2} \right) t^2 = 4\theta^2 t^2$$

$$x = \frac{-2t \frac{\theta}{n} + 2\theta t}{2} = \frac{n-1}{n} \theta t$$

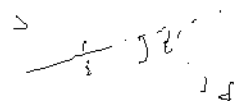
1.5. 证明光入射多层介质的出射方向仅与两端折射率有关

中间 $n \sin \theta = \cos \theta$. $i'_m = i_n$ $\square$.

1.6. 证明光线垂直入射光劈后, $\delta = (n-1)d$ ($d \ll 1$)

$$n \sin i = \sin i' \quad i = d \ll 1 \rightarrow ni = i'$$

$$\delta = i' - i \approx (n-1)d$$



1.7 由题 $\frac{OR}{OP} = \frac{n}{n'}$

在 $\triangle ORP$ 中 $\frac{OR}{OP} = \frac{\sin \angle RPO}{\sin (\pi - \angle PRO)} = \frac{\sin \angle RPO}{\sin \angle PRO}$

$$\text{又 } n \sin (\pi - \angle PRO) = n' \sin \angle RPO \quad \square$$

1.8. OP, RP 夹角 $\angle RPO = d$ 给定, $OP = n'$, $OR = OQ = n$

即求 $\angle ROQ = \delta$ 最小值.

$$RQ = 2n \sin(\frac{\delta}{2}) \rightarrow \text{求 } (RQ)_{\min}$$

因 $\angle RPO = d$ 给定 所以 $(RQ)_{\min} \rightarrow RP = RQ$ 即最小偏向角情形

$$RP^2 = n^2 + n'^2 - 2nn' \cos(\frac{\delta}{2})$$

$$\frac{RP}{\sin(\frac{\delta}{2})} = \frac{OR}{\sin(\frac{d}{2})} = \frac{n}{\sin(\frac{d}{2})} \quad \text{令 } nr = \frac{n'}{n}$$

$$\rightarrow \sqrt{1 + nr^2 - 2nr \cos(\frac{\delta}{2})} = \frac{\sin(\frac{\delta}{2})}{\sin(\frac{d}{2})}$$

$$nr^2 - 2r \cos(\frac{d}{2}) nr + 1 = \frac{\sin^2(\frac{\delta}{2})}{\sin^2(\frac{d}{2})} = 0$$

$$nr = \cos(\frac{\delta}{2}) + \sqrt{\cos^2(\frac{\delta}{2}) - 1 + \frac{\sin^2(\frac{\delta}{2})}{\sin^2(\frac{d}{2})}} = \cos(\frac{\delta}{2}) + \tan(\frac{d}{2})$$

$$= \frac{1}{\sin \frac{d}{2}} \left[\cos(\frac{\delta}{2}) \sin(\frac{d}{2}) + \sin(\frac{\delta}{2}) \cos(\frac{d}{2}) \right]$$

$$\sin(\frac{\delta+d}{2})$$

$$= \sin\left(\frac{\theta}{2}\right)$$

$$1.13 \quad n_1 \sin \theta_1 = n_2$$

$$\theta_1' = \frac{\pi}{2} - \theta_2 \quad n_0 \sin \theta_1 = n_1 \sin \theta_1'$$

$$\rightarrow \theta_1 = \arcsin \frac{n_1^2 - n_0^2}{n_0}$$

$$1.14 \quad n_g \sin i_0 \approx n \sin\left(\frac{\pi}{2}\right) = n$$

$$i = \frac{\pi}{2} - i_0 \quad n_g \sin i = \sin i'$$

$$\rightarrow n = n_g / \left(1 - \frac{(\sin i')^2}{n_g^2}\right) = (n_g^2 - \sin^2 i')'$$